

Ishan Mani Singh

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EDUCATION

Vellore Institute Of Technology

Chennai, Tamil Nadu

B.Tech in Computer Science and Engineering (Data Science) — CGPA: 8.85/10

2025 - 2029

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Go, JavaScript, Bash, TailwindCSS, Vanilla HTML

AI/ML Frameworks: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Hugging Face Diffusers, Stable-Baselines3, OpenCV, MediaPipe, Gymnasium, OpenAI

AI/ML: CNNs, Computer Vision, OCR, RAG, Reinforcement Learning, LLMs, LoRA Fine-Tuning

Backend & Data: FastAPI, Flask, Node.js, REST APIs, Redis, PostgreSQL, MongoDB, MySQL, SQLite

Systems & Tools: Docker, Linux, Git, GitHub, NumPy, pandas.

ACHIEVEMENTS

Top 2 FinTech domain, Top 10 Overall | Recursion Edition II TechnoVIT 2026 | [GitHub](#)

September 2026

- Secured **2nd place in the FinTech domain** and an **overall Top 10 position** among **550+ participants** in the flagship hackathon conducted by Microsoft Innovations Club, VIT Chennai.

Top 10 Finalist | IIT Madras E-Summit 2026

February 2026

- Secured a **Top 10 position across multiple events** at IIT Madras E-Summit 2026 while representing IEEE Computer Society, VIT Chennai.

VOLUNTEER EXPERIENCE

Microsoft Innovations Club (MIC), VIT Chennai

2025 – Present

AIML Department Technical Member

Chennai, India

- Contributed to the **AI/ML backend** of an AI-powered **Murder Mystery simulation** showcased to **200+ participants** at Pre-TechnoVIT 2025 and four subsequent Club Expo 2025 showcases | [GitHub](#).
- Contributed to **Spell Masters**, a 2D fantasy game showcased at Club Expo 2026, implementing **computer-vision gesture recognition** using 72-dimensional hand-landmark embeddings and cosine similarity for orientation-robust classification | [GitHub](#).
- Served as **main speaker** for **FormulAI**, a Formula One-themed RL workshop for **120+ students** at MIC's TechnoVIT 2026 AIML workshop, covering **PPO, reward engineering, environment design, and hyperparameter tuning** with Gymnasium and Stable-Baselines3 | [GitHub](#).

IEEE Computer Society, VIT Chennai

2025 – Present

Technical Member

Chennai, India

- Co-organized **PatchFest**, a technical open-source event with **100+ participants**, supporting participants in making and submitting open-source contributions.
- Supervised and reviewed participant **pull requests** and managed technical submissions using Git-based workflows, while supporting project development with Python and Node.js and other frameworks.

PROJECTS

AI Academic Assistant | [Project Page](#)

June 2025 – Present

- Collaborative AI academic assistant indexing **850+ PYQs across 170 courses** with **5 context-aware chat modes**, supporting document processing, RAG-based question answering, summarization, quizzes, and question generation.
- My Contribution:** Developed a **judge-based LLM layer** to classify incoming tasks and route them to appropriate AI workflows and model configurations.
- Implemented the **OCR processing layer** using **Tesseract and OpenCV** for scanned and handwritten academic material.
- Contributed to the **contextual RAG pipeline**, using contextual and sentiment-aware signals to improve task classification, retrieval, and response generation.

- Helped evaluate **LLM architectures and model-serving strategies**, including V-LLM and multiple model providers.

Layer 7 API Gateway & Distributed Rate Limiter | *Go, Redis, Lua, HTTP* | [GitHub](#) 2026 – Present

- Building a **Layer 7 reverse proxy and API gateway** in Go to protect upstream services through request filtering, traffic control, and load-balancing middleware.
- Implemented an **atomic Redis Lua token-bucket limiter** to handle concurrent requests without read-modify-write race conditions, with excess traffic rejected at the gateway using HTTP 429 responses.
- Engineered **client identity extraction and IP-spoofing protection**, preventing untrusted **X-Forwarded-For** headers from bypassing IP-based rate limits.
- Designed concurrent attack tests for **race-condition and header-spoofing scenarios** and validated the gateway before requests reach the protected upstream service.

NPC AI Middleware | [GitHub](#) 2025 – Present

- Designed an experimental **AI middleware architecture for game developers**, combining a backend service for AI workloads with a lightweight client-side integration layer for game engines.
- Designed a hybrid control system combining **algorithmic logic and LLM-driven behavior** for self-reliant control of NPCs and in-game objects.
- Implemented **LLM tool use** for structured world-state operations including persistent memory, character state changes, and in-game transactions.
- Built a **persistent NPC memory/state layer** with short-term conversational context, long-term memory, file locking, and asynchronous state updates.
- Engineered **JSON-constrained LLM outputs and recovery logic** to validate and repair model-generated tool calls before execution.

Digital Rubik's Cube Simulation | [GitHub](#) 2025 – Present

- Built a lightweight **Rubik's Cube simulation engine** as the environment for a planned Reinforcement Learning solver.
- Implemented **deterministic state transitions and compact state representation** to support repeated environment interaction and ML experimentation.
- Designed the engine as a foundation for **RL state, action, and reward modeling**, with the learning agent planned as a subsequent stage.
- Developed lightweight **C/Python implementations** to support simulation on constrained hardware when required.

Minecraft AI Agent Simulation | [GitHub](#) December 2023

- Developed an **LLM-driven autonomous agent** using Minecraft as a controllable 3D environment for experimenting with agent behaviour, natural-language interaction, spatial reasoning, and environment-grounded actions; conceptually aligned with research such as [Project Sid](#).
- Integrated **Mineflayer and Pathfinder** to provide environment perception and action primitives including block discovery, coordinate-based navigation, movement, and world interaction.
- Implemented an **event-driven asynchronous architecture** for handling environmental events, user commands, navigation tasks, and LLM inference concurrently.
- Integrated **Gemini 1.0 Pro** for contextual agent behaviour while maintaining deterministic execution of environment actions through explicit control handlers.

Generative AI & Local Model Development January 2023 – Present

- Building with **LLMs and local AI systems** since 2023, with hands-on work in local inference, context and memory management, streaming responses, structured outputs, tool use, and model configuration using Ollama and other Open Source tools.
- Worked with **LoRA-based fine-tuning**, including custom dataset preparation and adapter-based training; fine-tuned a **Stable Diffusion 1.5** model on a self-created dataset and have worked with countless Open Source tools such as ComfyUI, Automatic1111, Gradio, etc.